Pushing Medium (PM) — Consolidated Formula Sheet

March 2026

Core Assumptions

1. **Flat spacetime.** Space is a compressible physical medium on a fixed Euclidean background. No GR curvature, no Einstein equations, no horizons, no TOV.
2. **Scalar compression field.** Gravity arises from the scalar field ϕ with refractive index $n = e^{\phi}$. Sources: $\nabla^2\phi = -\frac{8\pi G}{c^2}\rho$.
3. **Density law.**
$$\rho = \rho_{\text{nuc}}\, e^{\phi}$$
4. **Force law for massive particles.**
$$\mathbf{a} = \tfrac{c^2}{2}\,\nabla\phi$$
(Half the photon coupling; recovers $-GM/r^2\,\hat{\mathbf{r}}$ exactly.)
5. **Light propagation.** Follows Fermat's principle in the optical metric:
$$ds^2_{\text{opt}} = -\frac{c^2}{n^2}\,dt^2 + (d\mathbf{r} - \mathbf{u}\,dt)^2$$
6. **Self-consistent EOS.**
$$P = \frac{c^2}{2}(\rho - \rho_{\text{nuc}})$$
7. **Stability cap.** The medium cannot be compressed beyond $\phi = 1$ ($n = e$, $\rho = e\,\rho_{\text{nuc}}$). This is the only strong-field limit in PM; it is not a compactness bound from GR.

Fields and Notation

Gravitational sector:

$$\phi(\mathbf{r},t) = \text{scalar compression field}, \quad \mathbf{u}(\mathbf{r},t) = \text{flow field}$$

Electromagnetic sector:

$$\psi(\mathbf{r},t) = \text{scalar EM field}, \quad \mathbf{w}(\mathbf{r},t) = \text{vector EM field}$$

Refractive index:

$$\boxed{n(\mathbf{r},t) = e^{\phi(\mathbf{r},t)}} \quad \Longleftrightarrow \quad \phi = \ln n$$

Static Gravity

Scalar field (Poisson):

$$\nabla^2\phi = -4\pi G_{\text{eff}}\,\rho_M$$

Point mass M:

$$\phi(r) = \frac{\mu_G M}{r}, \qquad n(r) = 1 + \frac{\mu_G M}{r} + \mathcal{O}(r^{-2})$$

Continuous distribution:

$$\phi(\mathbf{r}) = \mu_G \int \frac{\rho_M(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'$$

Vector flow field:

$$\nabla^2\mathbf{u} = -A_M\,\mathbf{J}_M$$

For a moving point mass with velocity $\mathbf{v}$:

$$\mathbf{u}(\mathbf{r}) = A_M M\,\frac{\mathbf{v}}{|\mathbf{r}-\mathbf{r}_M|}$$

Unified Source Structure

Scalar sources (gravity + EM + plasma):

$$S_\phi = 4\pi(G_{\text{eff}}\,\rho_M + K_E\,\rho_q + K_{\text{plasma}}\,n_e)$$

Vector sources (mass + charge currents):

$$\mathbf{S}_u = A_M\,\mathbf{J}_M + A_q\,\mathbf{J}_q$$

EM sources:

$$S_\psi = 4\pi K_E\,\rho_q, \qquad \mathbf{S}_w = B_q\,\mathbf{J}_q$$

Dynamic PM-Fluid Equations

Scalar field:

$$\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\,\mathbf{u}) = S_\phi + \frac{1}{c_\phi^2}\frac{\partial^2\phi}{\partial t^2}$$

Vector flow:

$$\frac{\partial\mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla\phi + \mathbf{S}_u + \frac{1}{c_u^2}\frac{\partial^2\mathbf{u}}{\partial t^2}$$

PM Lagrangian Density

Gravitational sector:

$$\mathcal{L} = \frac{(\partial_t\phi)^2}{2c_\phi^2} - \frac{|\nabla\phi|^2}{2} + \frac{|\partial_t\mathbf{u}|^2}{2c_u^2} - \frac{|\nabla\mathbf{u}|^2}{2} + \phi\,\nabla\cdot(\phi\mathbf{u}) + \phi\,S_\phi + \mathbf{u}\cdot\mathbf{S}_u$$

Electromagnetic sector:

$$\mathcal{L}_{\text{EM}} = \frac{(\partial_t\psi)^2}{2c_\psi^2} - \frac{|\nabla\psi|^2}{2} + \frac{|\partial_t\mathbf{w}|^2}{2c_w^2} - \frac{|\nabla\mathbf{w}|^2}{2} + \psi\,S_\psi + \mathbf{w}\cdot\mathbf{S}_w$$

Structurally identical with $(\phi,\mathbf{u}) \to (\psi,\mathbf{w})$.

Hamiltonian Formulation

Canonical momenta:

$$\pi_\phi = \frac{\partial_t\phi}{c_\phi^2}, \qquad \pi_{u_i} = \frac{\partial_t u_i}{c_u^2}$$

Hamiltonian density:

$$\mathcal{H} = \frac{c_\phi^2\pi_\phi^2}{2} + \frac{|\nabla\phi|^2}{2} + \frac{c_u^2|\pi_u|^2}{2} + \frac{|\nabla\mathbf{u}|^2}{2} - \phi\,\nabla\cdot(\phi\mathbf{u})$$

Kinetic and gradient terms are positive-definite; the system is stable in the linear regime.

Hamilton's equations:

$$\partial_t\phi = c_\phi^2\,\pi_\phi$$

$$\partial_t\pi_\phi = \nabla^2\phi + \nabla\cdot(\phi\mathbf{u}) + \phi\,\nabla\cdot\mathbf{u}$$

$$\partial_t u_i = c_u^2\,\pi_{u_i}$$

$$\partial_t\pi_{u_i} = \nabla^2 u_i + \partial_i(\phi^2)$$

Stress–Energy Tensor

Canonical form:

$$T^{\mu}{}_{\nu} = \sum_{\Psi\in\{\phi,\,u_i\}} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Psi)}\,\partial_\nu\Psi - \delta^{\mu}{}_{\nu}\,\mathcal{L}$$

Energy density T^{00}:

$$\mathcal{E} = \frac{(\partial_t\phi)^2}{2c_\phi^2} + \frac{|\nabla\phi|^2}{2} + \frac{|\partial_t\mathbf{u}|^2}{2c_u^2} + \frac{|\nabla\mathbf{u}|^2}{2} - \phi\,\nabla\cdot(\phi\mathbf{u})$$

Momentum density T^{0i}:

$$T^{0i} = \frac{\partial_t\phi\,\partial_i\phi}{c_\phi^2} + \frac{\partial_t u_j\,\partial_i u_j}{c_u^2}$$

Stress tensor T^{ij}:

$$T^{ij} = -(\partial_i\phi)(\partial^j\phi) - (\partial_i u_k)(\partial^j u_k) - \delta^{ij}\,\mathcal{L}_{\mathrm{PM}}$$

Conservation:

$$\partial_\mu T^{\mu\nu} = 0 \quad \text{(no external sources)}$$

Electromagnetic Field Equations

Scalar EM (electric-like):

$$\frac{\partial\psi}{\partial t} + \nabla\cdot(\psi\,\mathbf{u}) = S_\psi + \frac{1}{c_\psi^2}\frac{\partial^2\psi}{\partial t^2}$$

Static limit: $\nabla^2\psi = -4\pi K_E\,\rho_q$

Vector EM (magnetic-like):

$$\frac{\partial\mathbf{w}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{w} = -\nabla\psi + \mathbf{S}_w + \frac{1}{c_w^2}\frac{\partial^2\mathbf{w}}{\partial t^2}$$

Static limit: $\nabla^2\mathbf{w} = -B_q\,\mathbf{J}_q$

Effective fields:

$$\mathbf{E}_{\mathrm{eff}} = -\nabla\psi - \partial_t\mathbf{w}, \qquad \mathbf{B}_{\mathrm{eff}} = \nabla\times\mathbf{w}$$

Maxwell-like equations (linear, no-flow limit):

$$\nabla\cdot\mathbf{E}_{\mathrm{eff}} = 4\pi K_E\,\rho_q, \qquad \nabla\cdot\mathbf{B}_{\mathrm{eff}} = 0$$

$$\nabla\times\mathbf{E}_{\mathrm{eff}} = -\partial_t\mathbf{B}_{\mathrm{eff}}, \quad \nabla\times\mathbf{B}_{\mathrm{eff}} = \frac{\partial_t\mathbf{E}_{\mathrm{eff}}}{c^2} + B_q\,\mathbf{J}_q$$

Ray and Particle Kinematics

Ray position:

$$\frac{d\mathbf{r}}{ds} = c\,\hat{\mathbf{k}} + k_{\mathrm{Fizeau}}\,\mathbf{u} \qquad (k_{\mathrm{Fizeau}} = 1)$$

Ray direction:

$$\frac{d\hat{\mathbf{k}}}{ds} = \nabla_\perp\phi + \mathcal{F}(\hat{\mathbf{k}},\mathbf{u},\nabla\mathbf{u})$$

Massive particle acceleration (exact, all speeds):

$$\boxed{\mathbf{a}_{\mathrm{PM}} = +\tfrac{c^2}{2}\,\nabla\phi}$$

Light sees both metric components (g_{tt} and g_{rr}) → factor c^2. Massive slow particles see only g_{tt} → factor $c^2/2$. With $\phi = \mu_G M/r$: $\mathbf{a} = -GM/r^2\,\hat{\mathbf{r}}$ (Newtonian, exact).
Implementation: `core.massive_accel_medium`

Weak-field deflection angle:

$$\alpha_{\mathrm{PM}}(b) = \int_{-\infty}^{+\infty} \frac{-\mu_G M\,b}{n(r)\,r^3}\,dz \;\xrightarrow{n\to1}\; \frac{2\mu_G M}{b} = \frac{4GM}{c^2 b}$$

Moving-lens correction:

$$\alpha_{\mathrm{moving}} = \alpha_{\mathrm{static}}\Big(1 + \frac{v}{c}\Big)$$

Effective Metric

Optical metric (exact):

$$ds^2_{\mathrm{opt}} = -\frac{c^2}{n^2}\,dt^2 + (d\mathbf{r} - \mathbf{u}\,dt)^2$$

First-order expansion (small ϕ, small $\mathbf{u}$):

$$ds^2 = -(1+2\phi)c^2dt^2 - 2\,\mathbf{u}\cdot d\mathbf{r}\,dt + (1-2\phi)\,d\mathbf{r}^2$$

Metric components:

$$g_{tt} = -(1+2\phi), \qquad g_{rr} = (1-2\phi)$$

PPN parameters:

$$\gamma = 1, \qquad \beta = 1$$

Deformation Energy and Critical State

Landau–Ginzburg deformation energy:

$$U(\phi) = \varepsilon_0\Big(\phi^2 - \frac{\phi^3}{3}\Big), \qquad \varepsilon_0 = \rho_{\mathrm{nuc}}\,c^2$$

$$U'(\phi) = \varepsilon_0(2\phi - \phi^2), \qquad U''(\phi) = 2\varepsilon_0(1-\phi)$$

Stability criterion: $U''(\phi) > 0 \;\Leftrightarrow\; \phi < 1$

Critical point (onset of instability, $U'' = 0$):

$$\boxed{\phi_{\mathrm{crit}} = 1, \qquad n_{\mathrm{crit}} = e}$$

Density from field (exact):

$$\rho(\phi) = \rho_{\mathrm{nuc}}\,e^{\phi}$$

Self-consistent PM EOS (from $dP = \rho\frac{c^2}{2}d\phi$, $P\to0$ at $\rho\to\rho_{\mathrm{nuc}}$):

$$\boxed{P = \tfrac{c^2}{2}(\rho - \rho_{\mathrm{nuc}})}$$

Properties: $P(\rho_{\mathrm{nuc}}) = 0$; sound speed $c_s = c/\sqrt{2}$.

Critical thermodynamic values:

$$\rho_{\mathrm{crit}} = e\,\rho_{\mathrm{nuc}} \approx 6.25\times10^{17}\ \mathrm{kg\,m^{-3}} \;\approx 2.7\,\rho_{\mathrm{nuc}}$$

$$P_{\mathrm{crit}} \approx 1.87\times10^{34}\ \mathrm{Pa}$$

Implementation: `src/pushing_medium/critical_state.py`

Compact Star Structure (exact within PM)

PM uses flat space; no GR corrections at any compactness. Gauss's theorem on $\nabla^2\phi = -\mu_G \cdot 4\pi\rho$ gives the structure ODE system exactly:

ODE system (interior, $r > 0$):

$$\frac{dm}{dr} = 4\pi r^2\rho, \quad \frac{dP}{dr} = -\frac{Gm(r)\,\rho}{r^2}, \quad \frac{d\phi}{dr} = -\frac{\mu_G\, m(r)}{r^2}$$

Boundary conditions (at centre $r \to 0$):

$$m(0) = 0, \quad P(0) = \tfrac{c^2}{2}(\rho_c - \rho_{\rm nuc}), \quad \phi(0) = \ln(\rho_c/\rho_{\rm nuc})$$

Surface: integrate until $P = 0$ ($\rho \to \rho_{\rm nuc}$, $\phi \to 0$).

Hard stability cap:

$$\rho_c < \rho_{\rm crit} = e\,\rho_{\rm nuc} \quad \Leftrightarrow \quad \phi_c < 1$$

(From $U''(\phi_{\rm crit}) = 0$; medium cannot be compressed past this.)

Self-consistency: ϕ from Poisson $= \ln(\rho/\rho_{\rm nuc})$ from EOS, verified algebraically. Implementation: `src/pushing_medium/stellar_structure.py`

Surface Gravitational Redshift

PM formula (from optical metric $g_{tt} = -c^2/n^2$, conserved frequency):

$$\boxed{z_{\rm PM} = e^{\mu_G M/R} - 1}$$

Interior field ϕ integrates to zero at the surface; the physically relevant quantity is the *exterior* Poisson solution $\phi_{\rm ext}(R) = \mu_G M/R$, giving a ratio of emitted to received frequency $\nu_\infty/\nu_s = e^{-\phi_{\rm ext}} = n(R)^{-1}$.

GR Schwarzschild formula (for comparison):

$$z_{\rm GR} = \frac{1}{\sqrt{1 - 2GM/c^2R}} - 1$$

Weak-field expansion (both theories):

$$z_{\rm PM} \approx \frac{2GM}{c^2R}, \qquad z_{\rm GR} \approx \frac{GM}{c^2R}$$

PM predicts $\approx 2\times$ larger surface redshift than GR for the same (M, R). This is a direct, falsifiable prediction testable with X-ray spectroscopy of neutron stars.

NICER spot-checks (central values):

Source	M [$M_\odot$]	R [km]	$z_{\rm PM}$	$z_{\rm GR}$	ratio
J0030+0451	1.34	12.71	0.365	0.205	1.78
J0740+6620	2.08	12.39	0.642	0.408	1.57

EXO 0748−676 (Cottam 2002): $z_{\rm obs} \approx 0.35$, consistent with PM at NICER masses but insufficient to distinguish the theories without an independent (M, R) measurement.

Implementation: `stellar_structure.pm_surface_redshift`, `classical.gr_surface_redshift`, `scripts/surface_redshift_track.py`

GW Ringdown: PM vs GR

GR Schwarzschild QNM (l=2, n=0 fundamental; Leaver 1985):

$$f_{\rm QNM} = \frac{0.37367}{2\pi}\frac{c^3}{GM} \approx 12.1\,{\rm kHz} \times \frac{M_\odot}{M}$$

$$\tau_{\rm QNM} = \frac{GM}{2 \times 0.08896\, c^3} \approx 27.7\,\mu{\rm s} \times \frac{M}{M_\odot}, \qquad Q_{\rm GR} = \frac{\omega_R}{2\omega_I} \approx 2.10$$

(critically damped: energy absorbed by the horizon.)

PM: no horizon → no QNMs. The PM photon sphere lies at:

$$r_{\rm ps}^{\rm PM} = \mu_G M = \frac{2GM}{c^2} \quad \left(\tfrac{2}{3} \text{ of GR value } 3GM/c^2\right)$$

For realistic PM stars $R_\star > r_{\rm ps}^{\rm PM}$, so there are no exterior trapped photon orbits and no horizon QNMs.

PM surface (breathing) mode:

$$f_{\rm surf} = \frac{c_s}{2R_\star} = \frac{c}{2\sqrt{2}\,R_\star} \approx \frac{9.6\,{\rm kHz}}{R_\star/11\,{\rm km}}$$

$$\tau_{\rm surf} = \frac{Q}{\pi f_{\rm surf}}, \quad Q_{\rm PM} \gg Q_{\rm GR} \approx 2.1 \text{ (no horizon energy sink)}$$

PM surface echo delay:

$$\tau_{\rm echo} = \frac{2R_\star}{c} \approx 73\,\mu{\rm s} \times \frac{R_\star}{11\,{\rm km}}$$

PM maximum compact-star mass. Newtonian structure (no TOV); mass grows monotonically with ρ_c up to the stability cap:

$$\rho_c \le \rho_{\rm crit} = e\,\rho_{\rm nuc} \quad \Longrightarrow \quad M_{\rm max}^{\rm PM} \approx 13\text{–}14\,M_\odot$$

LIGO falsification summary:

- GW150914 ($29 + 36\,M_\odot$ progenitors): both exceed $M_{\rm max}^{\rm PM}$; PM predicts no compact progenitors.
- GW170817 ($1.46 + 1.27\,M_\odot$): within PM range, but PM predicts $R \approx 15$ km vs NICER ≈ 12 km (+30% offset).
- At $M \approx 2\,M_\odot$: $f_{\rm surf} \approx 8$ kHz $> f_{\rm QNM} \approx 6$ kHz (factor-of-1.3 difference; accessible to Einstein Telescope).

Impl.: `core.pm_surface_echo_delay`, `classical.gr_qnm_frequency`, `scripts/gw_ringdown_comparison.py`

Symbol	Value	Physical meaning
c	2.99792458×10^8 m/s	Speed of light
G	6.67430×10^{-11} m^3 kg^{-1} s^{-2}	Newton's constant
μ_G	$1.4861356 \times 10^{-27}$ m^2 kg^{-1} $= 2G/c^2$ (theory)	Grav. index coefficie
$k_{\rm TT}$	1.0 (exact)	GW power normalis
$k_{\rm Fizeau}$	1.0 (exact)	Moving-lens drag
A_M	$G/c^2 \approx 7.43 \times 10^{-28}$ m^3 kg^{-1} s^{-1}	Mass-current couplin
A_q	$< 3 \times 10^{-10}$ m^3 C^{-1} s^{-1}	Charge-current coup
κ_B	≈ 0 (ruled out)	Magnetic potential c
$\rho_{\rm nuc}$	2.3×10^{17} kg m^{-3}	Nuclear saturation d
Unconstrained: μ_E, K_E, B_q, $K_{\rm plasma}$, $\alpha_{\rm plasma}$, c_ϕ, c_u (assumed $\approx c$		

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